

Development of Driving Cycles for Galle

Heshani Rupasinghe and Terrence M Rengarasu
Department of Civil and Environmental Engineering
University of Ruhuna
Galle, SriLanka.

heshanirupasinghe@gmail.com and rengarasu@cee.ruh.ac.lk

Abstract—Driving cycle construction has become more popular around the world to estimate pollution emission and traffic analysis. This paper presents the traffic based analysis related to performance of B128 and B130 route in Galle, Sri Lanka using basic traffic engineering parameters such as travel time, average and maximum speed, average acceleration and deceleration. So the main objective of this paper is to identify traffic related parameters of above routes at Galle. The driving data was collected using on board measurement method by installing a prototype GPS tracking device in light-duty cars (AXIO, AQUA, FIT, Vitz). The cycle construction was done according to the segment based cycle construction method and roads were segmented considering the major intersections. Driving cycles were constructed for morning peak, noon peak, evening peak and off peak hours in B130 route and off peak hours in B128 route and the travel time, average speed, maximum speed, average acceleration and average deceleration were obtained. According to results, the maximum average speed of 27.53 km/h was obtained at off peak hours and travel time of 892s is taken to travel from Hapugala junction to Galle in noon peak hours. However the maximum speed was obtained at noon peak hours.

Keywords—Driving cycle, Route selection, Speed, Acceleration, Deceleration, Cycle length

I. INTRODUCTION

Driving cycle is a speed vs time profile which represents the driving behavior of a particular area. In different countries and organizations, driving cycles were developed to assess the vehicular performance through fuel consumption and pollutant emission and for traffic based analysis in different road ways. In the past driving cycle was developed to represent urban driving patterns using single speed- time trace [10], [16]. Driving cycle development was first initiated in early 70's in some regions of the world and US Federal test procedure 72 and US federal test procedure 75 were considered as first driving cycles developed in the world [17]

Data on speed vs time is very important for the traffic engineers to determine the average speeds for a particular road and for the designing and operation of the road. In early 50s speed and time profile concept was developed by the National research council and the Highway research board to assemble and analyze data on motor vehicle speed distribution under the different conditions of physical, environment and traffic conditions [1]. Then with the time driving cycle development concept was introduced as an advanced figure of speed time

profile in Los Angeles to determine the contribution of auto mobiles to photochemical smog in late 50s [10].

In early 70's the development of driving cycle was rapidly increased with different construction methods. The FTP 72 and FTP 75 cycles were developed in early 70's by using whole test run data and they are considered as the 1st driving cycles developed [13].

Standard driving cycles have wide range of uses. FTP 72, FTP 75, Sydney driving cycle, driving cycle for Chinese cities and driving cycle for Pune, India are some of the cycles derived to estimate emission based analysis [13], [12], [22], [11]. European driving cycles developed such as IMC and Hong Kong driving cycles developed were mainly targeted the traffic based analysis [14], [20].

Driving cycle development is becoming more and more popular through out the world with the rapid increment of vehicles. In between 2010 to 2013 there were 16% increment in vehicular population according to World Health Organization (WHO). At the present, industry analysis shows that already there are 1.2 billion vehicles in the world. Navigate research estimates in a new forecast that light duty vehicles make upto 95% of that in total. There by in 2035 number of vehicular population will be 2 billion all over the world [9]. Therefore it has a higher importance in developing driving cycles for the purpose of estimating traffic and the emission.

Although there are many driving cycles developed in world-wide for different cities such as Hong Kong [20], Bangkok [19], Pune India [11], China [3], Sydney [12], USA [1], Colombo [5] and for SEW in Sri Lanka [6] etc., no study has done yet to determine a driving cycle for Galle city. Therefore here it focuses on determining a driving cycles for the Galle area. The objectives of the research is to identify the driving characteristics such as average speed, acceleration and deceleration around Galle subways and to determine average cycle length.

II. METHODOLOGY

The Fig. 1 shows the process of cycle construction adopted in developing driving cycle for Galle.

A. Construction of GPS tracking device

Construction of GPS tracking device was the first step in constructing the driving cycle. Key components includes are Arduino UNO board, GPS sensor and External memory.

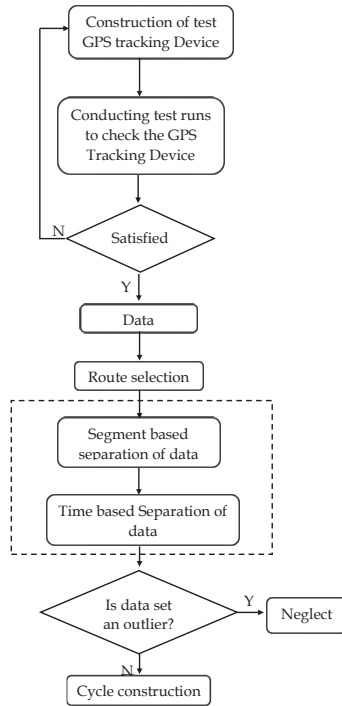


Fig. 1. Cycle construction flow chart

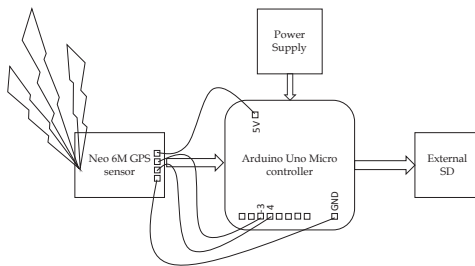


Fig. 2. Sketch of constructed GPS device

Sketch of the constructed instrument is shown in Fig. 2 and it was coded using Arduino programming.

B. Data Collection

According to the past literature related to the driving cycle development, data collection for the driving cycle was done in three ways. They are On board measurement method, chase a car method and hybrid method [7], [21]. Even though the chase a car method requires less number of resources and less cost, the main problem to apply this method to Galle, Sri Lanka is that it is very dangerous to follow a car due to aggressive driver behavior. Also accuracy of the data is less as the delays occur due to driver response time and accelerations and decelerations can be over estimated. In on

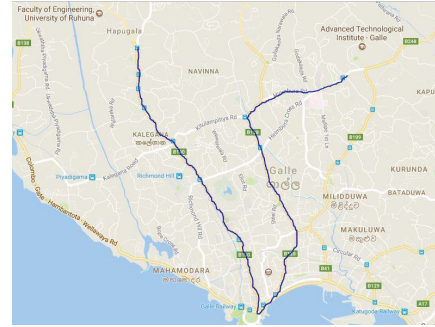


Fig. 3. Selected routes

N

board measurement method, data accuracy is higher and it gives the real world driving data. But it requires higher cost. Due to that to measure and record the real world driving data, on board measurement method is used. It is more accurate, reliable, over estimations and lower estimations are not present and the delays due to driver responses are not present. Data is collected by installing the instrument in Private vehicles of lecturers in Faculty of Engineering, University of Ruhuna only. Test vehicles include light duty cars AQUA, Vitz, AXIO and FIT. Constructed GPS tracking device was powered using the lighter of the car and device starts to collect data once the car switch on and device stops data collection once the car switched off. Cars were allowed to travel on B130 and B128 routes as shown in Fig. 3 and total data collection process was carried for one and half months. Total length of B130 route was 6 km from Galle to Hapugala and B128 was 7 km from Galle to Karapitiya .

C. Cycle construction

There are four basic cycle construction methods available to construct driving cycles. They are

- Micro trip based cycle construction: Micro trip is defined as the speed Vs time profile between two adjacent stops, including leading idling period [8]. More fuel is consumed when vehicle under go "stop-go" conditions. Traffic signal congestion generate most of the stop-go conditions and lead to higher fuel consumption [3]. So this is a better approach for emission related driving cycle construction.
- Segment based cycle construction: This is also same as micro trip based cycle construction. But instead of adjacent stops, road segments were selected [4]. Since the segments are separated according to the traffic behaviors, this method is more suitable for traffic based cycle constructions.
- Cycle construction with pattern based classification: This is done by dividing whole trip in to kinematic sequences using statistical methods.
- Modal cycle construction: In this method all the real world driving data is divided into accelerations, deceleration, cruising and idling components. Markov Chain theory is used in the construction assuming

TABLE I. ACTUAL NUMBER OF TRIP SEGMENTS MADE

Location and Time	HK	KJ	JM	MG	
MP	9	2	2	2	
NP	7	7	4	4	
EP	5	6	4	4	
OP	18	22	16	11	
Location and Time	GEMS	EMSS	SB	BK	KK
OP	1	2	9	9	5

Note: Hapugala junction to Kalegana Junction (HK), Kalegana Junction to Julgaha Junction (KJ), Julgaha Junction to Minuwangoda Junction (JM), Minuwangoda Junction to Galle (MG), Galle to End of main street (GEMS), End of main street to Sarenthukade Junction (EMSS), Sarenthukade Junction to Beligaha Junction (SB), Beligaha Junction to Kahaduwaitha Junction (BK), Kahaduwaitha Junction to Karapitiya (KK).

that likelihood of particular modal event lies only on the previous modal event [15].

After considering all the four cycle construction methods, segment based cycle construction method was selected as the best method to construct driving cycle as the objective of the cycle construction is traffic based analysis. Traffic engineering purpose driving cycle should represent the driving behavior of each location of the route. So the routes were segmented according to the number of intersections and cycles were separately constructed for morning peak (MP) from 0700 to 0900, noon peak (NP) from 1200 to 1400, evening peak (EP) from 1600 to 1800 and off peak (OP) hours. As the GPS sensor records the data in UTC (Universal coordinated time) format, morning peak is from 0130 to 0230, noon peak is from 0630 to 0830 and evening peak is from 1030 to 1230. The Actual trip segments made for the data collection were summarized in Table I.

D. Outliers analysis

The time taken to travel from one segment to another is differs from trip to trip as the traffic behaviors are not unique. The days which were having excessive traffic in each segment was removed by performing outliers analysis with the help of Equation numbers 1, 2 and 3.

Q1 = First Quantile

Q3 = Third Quantile

$$IQR = Q3 - Q1 \quad (1)$$

$$UB = Q3 + 1.5 * IQR \quad (2)$$

$$LB = Q1 - 1.5 * IQR \quad (3)$$

The time values which were not in between the lower bound and the upper bound were considered as the outliers. Outlier data sets were removed when constructing the driving cycle.

E. Limitations

- Limitations in GPS device construction

The construction cost of one GPS device is around Rs. 5000/=. So that the no of GPS device that can construct with the given amount of money was limited to four GPS devices.

- Limitations in data collection

The main limitation occurred in data collecting was the reliability. As the no of GPS devices are Limited and the storage of the GPS device is limited, data needed to be collected within every week. So that the instrumented vehicles should have the easy access. Therefore the GPS devices were installed only in the vehicles which are inside the faculty. The other main limitation in data collection was the travel route. To represent the actual traffic behavior, routes can't be predetermined. On the other hand, limited no of vehicles can't cover up all the routes in Galle area. Even the cycle construction requires morning, noon, evening and off peak hours the data on peak hours were lesser than the off peak hours.

- Limitations in route selection

In constructing driving cycles, route selection was done mainly based on AADT, route type and land use [19], [12], [21]. But in constructing driving cycle for Galle, route selection had to be done according to the availability of the data.

- Limitations in cycle construction

As the driving cycle was developed for the traffic engineering purposes, cycle construction was done according to segment base cycle construction method. Route segmentation can be done either constant distances [2], or according to the availability of intersections [18]. Accuracy of the developed cycle is higher when the segment become smaller. When constructing cycle depending on the intersections, it shows the accurate traffic at the intersection. But the accuracy of the middle is lesser. As the manual segmentation was used with the help of fusion table, constructing driving cycle based on constant distances was a major limitation. As the GPS device does not give the exact distance based speed, time and location data constant distances based segmentation was not possible. So that intersection based cycle construction method had to be followed even though it is less accurate.

III. RESULTS AND DISCUSSION

The segments of the roads were added together and final averaged speed time data were plotted in the graphs and average speed, maximum speed, average acceleration and average deceleration was obtained.

According to the Fig. 4, four segments of the chart indicates the driving behavior of road sections Hapugala Junction to Kalegana Junction, Kalegana Junction to Julgaha Junction, Julgaha Junction to Minuwangoda Junction and Minuwangoda Junction to Galle at morning peak hours. At each intersection the speed goes below 10 km/h and near Sangamiththa Girls school the speed goes below 10 km/h. There are many sudden accelerations and decelerations and it clearly depicts a traffic congestion at morning peak hours.

The total time taken to travel from Hapugala Junction to Galle was 840 s (14 minutes). The average speed, maximum

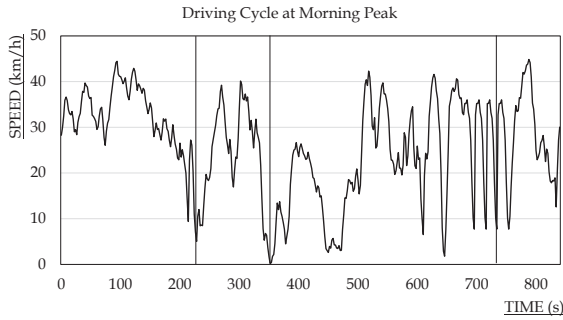


Fig. 4. Driving cycle at morning peak hours in B130 route

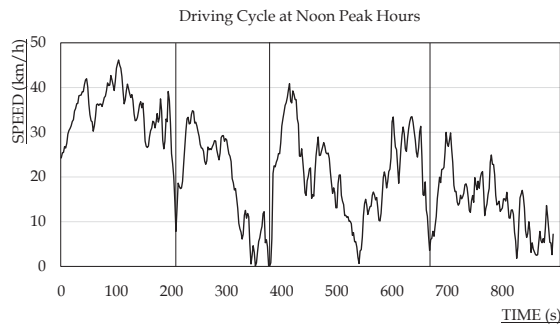


Fig. 5. Driving cycle at noon peak hours in B130 route

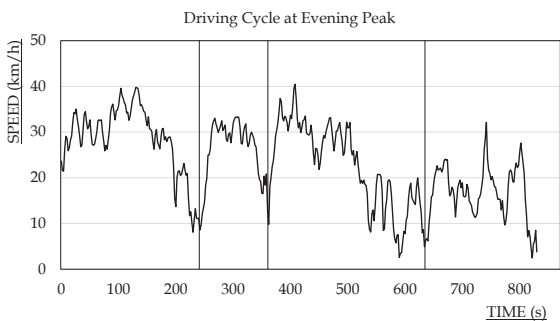


Fig. 6. Driving cycle at evening peak hours in B130 route

speed, average acceleration and average deceleration be 26.09 km/h, 44.85 km/h, 1.39 km/h² and 1.22 km/h² respectively.

According to the Fig. 5, Noon peak hours driving cycle took more time to reach Galle from Hapugala Junction and time taken is 892 s (15 minutes). The speed at Sangamiththa Girls school reached nearly to zero due to higher traffic condition at noon peak hours. The maximum speed of 46.151 km/h is obtained in Hapugala Junction to Kalegana Junction segment. The average speed of the total driving cycle is 22.00 km/h, which records the minimum average speed of all four driving cycles. The average acceleration is 0.877 km/h² and average deceleration is 1.018 km/h².

According to the Fig. 6, evening peak hours has less traffic congestion compared to morning peak and noon peak hours. Speed variations are huge only at the intersections and near

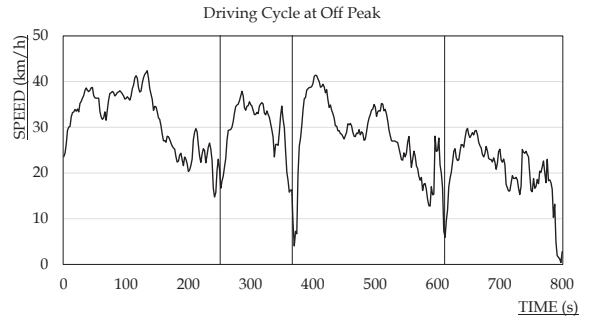


Fig. 7. Driving cycle at off peak hours in B130 route

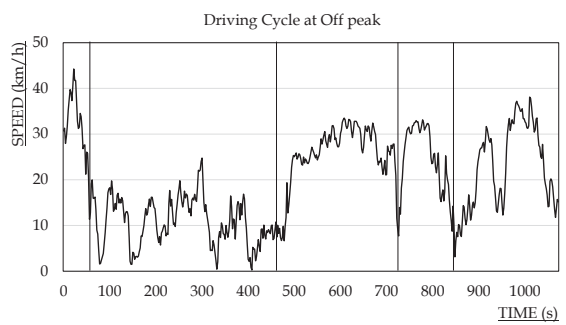


Fig. 8. Driving cycle at off peak hours in B128 route

to Minuwangoda junction it creates a lot of traffic. The total time taken to travel from Hapugala Junction to Galle is 830 s (13.8 minutes). The average speed is 27.53 km/h and maximum speed of 42.396 km/h records in between Julgaha junction to Minuwangoda junction. The average acceleration and average deceleration are 1.65 km/h² and 1.91 km/h² respectively.

According to the Fig. 7, the off peak hours driving cycle shows the lowest traffic congestion. The time taken to travel from Hapugala junction to Galle is 800 s (13.33 minutes) which is the lowest from all four travel times. Also the highest average speed of 27.53 km/h records in the off peak hours. Even though the speed reduces near junctions as usual. The average acceleration and average decelerations are 0.688 km/h² and 0.698 km/h² respectively.

The driving cycle for B128 route in off peak hours are shown in Fig. 8. The road segments are decided according to the intersections and they are Galle to End of main street, End of main street to Sarenthukade junction, Sarenthukade junction to Beligaha junction, Beligaha junction to Kahaduwattha junction and Kahaduwattha junction to Karapitiya. Even though the Fig. 8 shows driving cycle at off peak hours, End of main street to Sarenthukade junction segment is having more traffic while other road segments show less traffic. This is due to the narrow road width and there are many shops along the road. Due to customers vehicle parking along the both sides of the road, the road width have become more narrower.

According to the Fig. 8, the total travel time is 1072 s (18 minutes). The average speed is 19.67 km/h and the maximum speed is 44.251 km/h. The average acceleration and average

TABLE II. SUMMERY OF THE RESULTS

Cycle	Time (s)	Average Speed (km/h)	Maximum Speed (km/h)	Average acceleration (km/h ²)	Average deceleration (km/h ²)
MPB130	840	26.09	44.85	1.39	1.22
NPB130	892	22	46.15	0.88	1.02
EPB130	830	23.53	40.47	1.65	1.91
OPB130	800	27.53	42.4	0.69	0.69
OPB128	1072	19.67	44.25	0.87	0.87

deceleration are 0.86 km/h² and 0.87 km/h² respectively.

IV. CONCLUSION

The constructed driving cycle for B130 route from Hapugala junction to Galle in three main peak hours and off peak hours revealed that the maximum time taken to travel from Hapugala junction to Galle is 892 seconds. It can be round off to 15 minutes. Maximum speed for the road was found to be 46.15 km/h. Maximum average speed was 27.53 km/h at off peak hours. Maximum acceleration was 1.65 km/h² and maximum deceleration was 1.91 km/h².

Driving cycle for the B128 route from Galle to Karapitiya shows the average cycle length as 1072 seconds. It can be round off to 18 minutes. Average speed, maximum speed, average acceleration and deceleration was 19.67 km/h, 44.25 km/h, 0.87 km/h² and 0.87 km/h² respectively.

As the B128 and B130 routes are having different number of intersections and different physical conditions, traffic parameters obtained from off peak hours in both routes are varied. But maximum average speed of a road implies the overall road condition and time length for a journey which should be similar or identical to any given B class road having same no of intersections, distances and land use pattern. Maximum speed implies the optimum road condition and percentage time for acceleration and deceleration of the road and it too can be used for either similar or identical B class route with similar physical road characteristics.

When considering the road segments, developed driving cycles can be directly applied to analyze traffic engineering parameters in identical segments within a given two intersections.

The developed cycle construction methodology is relatively low cost and easy to follow. So this can be used as a good reference to the traffic based analysis in similar routes.

ACKNOWLEDGMENT

The authors would like to acknowledge Dr. Terrence M Ren-garasu, Supervisor of this research project for his continuous guidance and supervision and Dr. W.K.M.R.T.W. Bandara, Dr. Ruwan Gallage and Mr. Kasun Wimalasena for the help given in collecting data.

REFERENCES

- [1] Berry, S. Donald and Belmont, M. Daniel and others, "Distribution of vehicle speeds and travel times" *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, The Regents of the University of California, 1951.
- [2] Braunsteins, Janis and Watson, HC and Milkins, EE, "Melbourne peak cycle", 1981.
- [3] Chen, Changhong and Huang, Cheng and Jing, Qiguo and Wang, Haikun and Pan, Hansheng and Li, Li and Zhao, Jing and Dai, Yi and Huang, Haiying and Schipper, Lee and others, "On-road emission characteristics of heavy-duty diesel vehicles in Shanghai" *Atmospheric Environment*, Vol. 41, No.26 pp.5334-5344, 2007.
- [4] Dai, Zhen and Niemeier, Deb and Eisinger, Douglas, "Driving cycles: a new cycle-building method that better represents real-world emissions" Department of Civil and Environmental Engineering, University of California, Davis, 2008.
- [5] Gamalath, I., Fernando, C., Galgamuwa, U., Perera, L. and Bandara, S., "Methodology to develop a driving cycle for a given mode and traffic corridor; Case study for Galle Road, Colombo, Sri Lanka." 2014.
- [6] Galgamuwa, U., Perera, L. and Bandara, S., "A Representative Driving Cycle for the Southern Expressway Compared to Existing Driving Cycles." *Transportation in Developing Economies*, Vol. 2(2), pp.22. 2016.
- [7] H.Y Tong and W.T. Hung and C.S. Cheung, "On road motor vehicle emissions and Fuel consumption in urban driving conditions" *Air and waste management association*, pp.543-554, 2000.
- [8] Ho, Sze-Hwee and Wong, Yiik-Diew and Chang, Victor Wei-Chung, "Developing Singapore Driving Cycle for passenger cars to estimate fuel consumption and vehicular emissions" *Atmospheric Environment*, Vol. 97, pp.353-362, 2014.
- [9] John Voelcker, "Green Car Reports" 2014.
- [10] K.A. Stonex, "Survey of Los Angeles traffic characteristics", *U.S. Highway research board*, pp.509-538, 1972
- [11] Kamble, Sanghpriya H and Mathew, Tom V and Sharma, GK, "Development of real-world driving cycle: Case study of Pune, India" *Transportation Research Part D: Transport and Environment*, Vol.14, No.2, pp. 132-140, 2009.
- [12] Kent, JH and Allen, GH and Rule, G, "A driving cycle for Sydney" *Transportation Research*, Vol.12, No.3, pp. 147-152, 1978.
- [13] Kruse, RE and Huls, TA, "Development for the federal urban driving cycle" *SAE paper*, Vol.730553, pp. 7, 1973.
- [14] Kuhler, M and Karstens, D, "Improved driving cycle for testing automotive exhaust emissions" *SAE Technical paper*, 1978.
- [15] Lin, Jie and Niemeier, DA, "Regional driving characteristics, regional driving cycles" *Transportation Research Part D: Transport and Environment*, Vol. 8, No. 5, pp.361-381, 2003.
- [16] Lyons, TJ and Kenworthy, JR and Austin, PI and Newman, PWG, "The development of a driving cycle for fuel consumption and emissions evaluation" *Transportation Research Part A: General*, Vol 20, No.6, pp.447-462, 1986.
- [17] Quarles, J, "Control of air pollution from new motor vehicles and new motor vehicle engines. Importation of catalyst-equipped vehicles" *Federal Register of United States*, Vol 39, No.204, 1974.
- [18] Smith, Laron and Beckman, Richard and Anson, Doug and Nagel, Kai and Williams, Michael, "TRANSIMS: Transportation analysis and simulation system" *Los Alamos National Lab., NM (United States)*, 1995.
- [19] Tamsanya, Sukanya and Chungpaibulpattana, Supachart and Atthajariyakul, Surat, "Development of automobile Bangkok driving cycle for emissions and fuel consumption assessments" *Second Joint International Conference on Sustainable Energy and Environment, Bangkok*, 2006.
- [20] Tong, HY and Hung, WT and Cheung, CS, "Development of a driving cycle for Hong Kong" *Atmospheric Environment*, Vol 33, No.15, pp.2323-2335, 1999.

- [21] Tong, HY and Hung, WT, "A framework for developing driving cycles with on-road driving data" *Transport reviews*, Vol 30, No.5, pp.589–615, 2010.
- [22] Wang, Qidong and Huo, Hong and He, Kebin and Yao, Zhiliang and Zhang, Qiang, "Characterization of vehicle driving patterns and development of driving cycles in Chinese cities" *Transportation research part D: Transport and Environment*, Vol 13, No.5, pp. 289-297, 2008.